

PROJECT DESIGN PHASE

Project Name: Script-Controlled ACL – Restrict Record Access Based on Field Value

Module 1: Problem Solution Fit

1. Problem Statement

In enterprise ServiceNow applications, sensitive records may need to be visible only to users who satisfy specific role and field-based conditions. If access is controlled only through general table permissions or roles, users may be able to view records that belong to a different branch. This creates important record-level security challenges:

Unauthorized Record Visibility: Users without the required access should not be able to view restricted branch records.

Field-Based Access Requirement: Access must be controlled using the value of the Branch field, with EEE records restricted to the intended users.

Inconsistent Access Control: Role-only security does not fully express the required record-level condition when access depends on record data.

Administrative Access Requirement: Administrators must retain full access to records regardless of the branch value.

2. Target Audience & Stakeholders

ServiceNow System Administrators, Application Administrators, Security Administrators, and developers responsible for implementing and maintaining record-level access controls.

3. Problem-Solution Fit Matrix

Identified Pain Points	Proposed System Capability	Business Value & Outcome
Unauthorized users can view restricted records	Script-controlled READ ACL	Prevents users who do not satisfy the required access condition from viewing protected records.
Access depends on a record's Branch value	Data condition: Branch is EEE	Restricts the READ access rule to records whose Branch field is EEE.
Role-only security may not provide the required record-level control	Custom role requirement using bb1	Ensures the intended user role is part of the access decision.
Administrators need unrestricted visibility	Script check using gs.hasRole('admin')	Allows administrator users to retain full access to records regardless of branch.
Different operations require different permissions	Separate READ, CREATE, WRITE and DELETE ACLs with roles bb1–bb4	Provides controlled permissions for viewing, creating, editing and deleting records.

4. Summary of Fit

The proposed solution directly addresses record-level security requirements by combining ServiceNow ACLs, custom roles, field-based conditions, and script-controlled access decisions. The READ ACL uses the Branch field condition and role-based checks to restrict access to EEE records, while administrators retain full access. Separate CREATE, WRITE, and DELETE ACLs extend controlled permissions to record creation, modification, and deletion. This provides a structured approach to protecting sensitive records from unauthorized viewing and changes.